Ballistic Pendulum

Equipment:

1	Rotary Motion Sensor	PS-2120
1	Mini Launcher	ME-6825A
2	Photogate Heads	ME-9498A
1	Photogate Mounting Bracket	ME-6821A
1	Mini Launcher Ballistic Pendulum	ME-6829
1	45 cm Steel Rod	ME-8736
1	Large C Clamp (need one only)	ME-7285
1	550 Universal Interface	UI-5001
1	PASCO Capstone	UI-5400

Introduction:

A ballistic pendulum is used to determine the muzzle velocity of a ball shot out of a Projectile Launcher. The laws of conservation of momentum and conservation of energy are used to derive the equation for the muzzle velocity.

Theory:

The ballistic pendulum has historically been used to measure the launch velocity of a high speed projectile. In this experiment, a projectile launcher fires a steel ball (of mass m_{ball}) at a launch velocity, v_o. The ball is caught by a pendulum of mass m_{pend}. After the momentum of the ball is transferred to the catcher-ball system, the pendulum swings freely upwards, raising the center of mass of the system by a distance h. The pendulum rod is hollow to keep its mass low, and most of the mass is concentrated at the end so that the entire system approximates a simple pendulum. During the collision of the ball with the catcher, the total momentum of the system is conserved. Thus the momentum of the ball just before the collision is equal to the momentum of the ball-catcher system immediately after the collision:

$$m_{ball}v_o = Mv \qquad \text{Eq. 1}$$

where v is the speed of the catcher-ball system just after the collision, and

$$M = m_{ball} + m_{pend} \qquad \text{Eq. 2}$$

During the collision, some of the ball's initial kinetic energy is converted into thermal energy. But *after* the collision, as the pendulum swings freely upwards, we can assume that energy is conserved and that all of the kinetic energy of the catcher-ball system is converted into the increase in gravitational potential energy.

$$\tfrac{1}{2} Mv^2 = Mgh = MgL(1-\cos\theta) \qquad \text{Eq. 3}$$

where g =9.80665 m/s^2, and the distance h is the vertical rise of the center of mass of the pendulum-ball system.

Combining equations (1) through (3), to eliminate v, yields

$$v_o = \frac{m_{ball} + m_{pend}}{m_{ball}} \sqrt{2gh} \qquad \text{Eq. 4}$$

and from Figure 1 we have

$$h = L\,(1-\cos\theta), \qquad \text{Eq. 5}$$

with L being the distance between the center of mass and the axis the rod rotates around, and θ is the angle the rod rotates through before stopping.

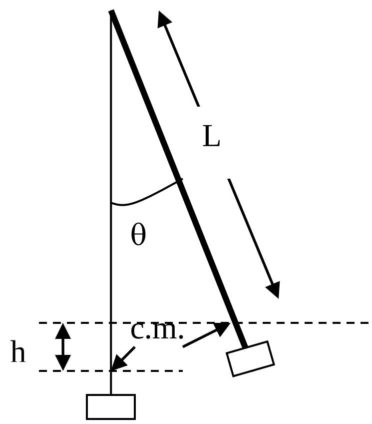

Figure 1: Finding the height

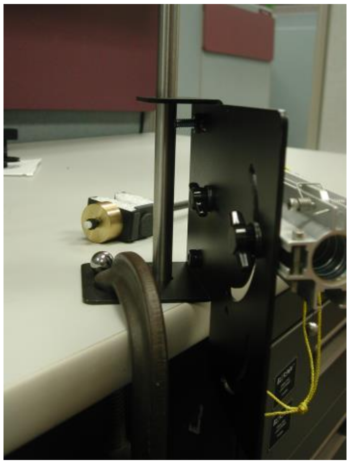

Figure 2: Attaching the Launcher

Figure 3: Full Setup

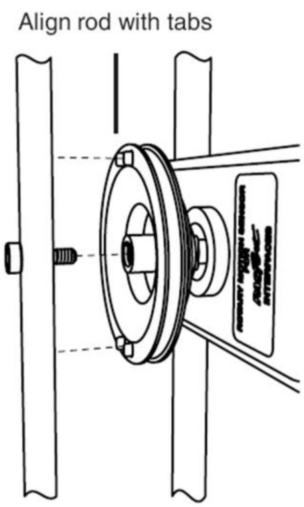

Fig. 4: Attaching the Pendulum

Fig. 5: Catcher Height

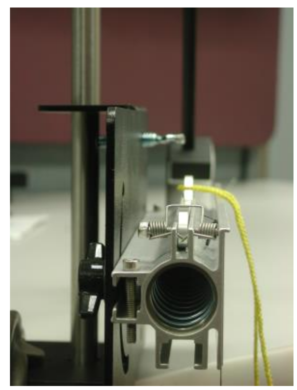

Fig. 6: Align the Pendulum

Setup:

1. Attach the 100g ballast mass to the bottom of the catcher.

2. Attach the mini launcher to the bracket using two thumbscrews and the hole and slot shown in Figure 2. Clamp the bracket to a table as shown in Figure 2. Put the 45 cm rod throught the two holes in the bracket and secure with the round black thumbscrew.

3. Adjust the mini launcher for an angle of 0^0. Tighten the thumbscrews.

4. Attach the Rotary Motion Sensor (RMS) to the top of the 45 cm rod as in Figure 3. Plug the RMS into any *PasPort* input on the 550 Universal Interface. If the side of the Rotary

Motion Sensor *without* the model number on the label is facing you, the angles you read will be positive. (If the Rotary Motion Sensor is mounted the other way, it will measure negative angles, but it really doesn't matter. Just ignore any minus signs.)

5. Slide the three-step pulley onto the Rotary Motion Sensor shaft with the largest pulley facing out.

6. Attach the pendulum to the Rotary Motion Sensor using the hole near the end of the pendulum and the long silver thumbscrew that came with the pendulum. Align the pendulum rod with the plastic tabs on the pulley so it cannot slip on the pulley. See Figure 4.

7. Adjust the position of the Rotary Motion Sensor so the pendulum is aligned with the launcher as shown in Figure 5. The center of the notch in the Styrofoam catcher should be at the same height as the center of barrel of the launcher. The pendulum should almost touch the launcher, but not quite (leave a 1 mm gap or so.) It is important that the plane of the pendulum be aligned with the axis of the launcher. Sight down the launcher and swing the pendulum back to check as in Figure 6. Be careful not to foul up the alignment when loading the launcher.

Procedure:

1. To load the launcher, swing the pendulum out of the way, place the ball in the end of the barrel and, using the pushrod (a plastic rod in your toolbox) included with the launcher, push the ball down the barrel until the trigger catches in the third position (you need to feel the click).

2. Return pendulum to its normal hanging position. Wait until it stops moving.

3. Click RECORD.

4. Launch the ball so that it is caught in pendulum.

5. After the pendulum has swung out and back, click STOP.

6. Repeat for a total of five times.

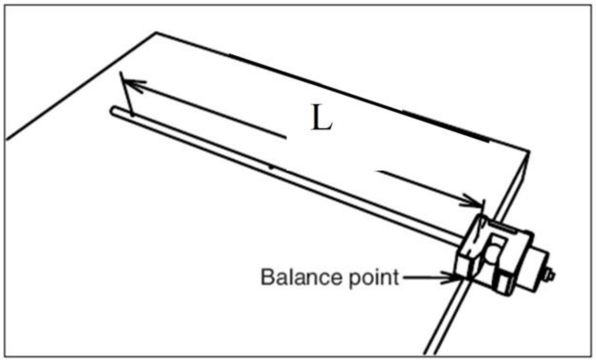

Figure 7: Finding the Center of Mass

Find the Mass and Center of Mass:

1. With the ball still in the catcher, remove the pendulum from the Rotary Motion Sensor.

2. Remove the screw from the pendulum shaft.

3. With the ball still in the catcher, place the pendulum at the edge of a table with the pendulum shaft perpendicular to the edge and the counterweight hanging over the edge. Push the pendulum out until it just barely balances on the edge of the table. The balance point is the center of mass. (See Figure 7)

4. Measure the distance, L, from the center of rotation (where the pendulum was attached to the Rotary Motion Sensor) to the center of mass. Click open the Calculator at the left of the screen. Enter your value for L in line 3 (replace my value of 0.356 m).

5. Remove ball from the catcher.

6. Measure the mass of the pendulum (without the ball). Record this value in line 6 of the Calculator (replace my value of 153.6 g).

7. Measure the mass of the ball. Record this value in line 5 of the Calculator (replace my value of 16.5 g).

Analysis:

1. Click on the black triangle by the Run Select icon on the graph toolbar and select Run #1.

2. Click on the Scale to Fit icon on the graph toolbar.

3. Click on the black triangle by the Statistics icon and select Maximum if your angles are positive or Minimum if your angles are negative. Click anywhere to close the selection box. Click the Statistics icon. The maximum value for the angle should appear (ignore any minus sign.)

4. Enter the maximum angle value in the "max angle" column on the Initial Ball Speed from Equation 4 table in the first row.

5. Repeat for Run #2 through Run #5, entering the values in rows 2 – 5.

6. Click open the calculator and verify that the values for "h" and "v zero" in the Initial Ball Speed table are being calculated from Equation 5 (see line 4 in the Calculator) and Equation 4 (see line 7 in the Calculator) from Theory. Click the Calculator closed.

7. The values in the two boxes below are the average value for v_0 and the uncertainty in the value calculated from the standard deviation of the five values in the table.

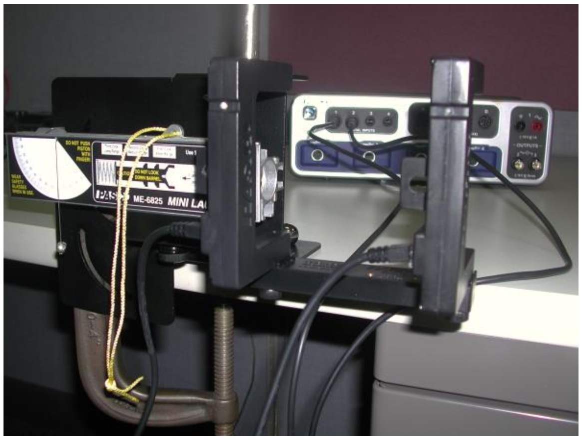

Figure 8: Photogate Setup

Verifying the Initial Speed of the Ball Using Photogates

1. Remove the pendulum from the Rotary Motion Sensor so it does not hang in front of the launcher.

2. Attach two photogates to the photogate bracket. To mount the bracket to the launcher, put a square nut onto the bracket and slide the bracket into bottom T-slot of the barrel. Slide the bracket back until the photogate is as close to the barrel as possible without blocking the beam (see Figure 8). Tighten the thumb screw to secure the bracket in place.

3. Attach the photogate closest to the launcher to Digital Input 1 on the 550 Universal Interface. Attach the other photogate to Digital Input 2.

4. Click PREVIEW at the lower left of the screen. The clock will begin to run.

5. Load the ball and cock the launcher to the third position. Pass your hand through the second photogate (farthest from the launcher) to arm the timer (You only need to pass once for each loading).

6. Pull the cord to launch the ball. Be careful that nobody is in the firing range. You might block the ball with a plastic waste basket. A green number should appear in the first row of the Initial Speed table. If it is around 5 m/s, click Keep Sample (lower left of screen) to accept it. The number will turn black and a green number will appear in line 2 of the table. If the number is around 0 m/s, you did not arm the timer correctly and should repeat steps 5 and 6 without accepting the number.

7. Repeat steps 5 and 6 until you have five numbers (should be nearly the same) in the Initial Speed table.

8. The two boxes below give the average value for v_0 measured by the Photogates and the uncertainty in the average due to the standard deviation of the values in the table.

Conclusions:

How well does the initial speed, v_0, calculated from Equation 4 agree with the value measured directly using the photogates? What does this show? Why is error analysis important?